

LANGRES, France

WMO: 072830

Lat: 47.8436N

Lon: 5.3375E

Elev: 1532

StdP: 13.90

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.1	21.4	11.9	10.7	22.3	16.5	13.4	24.2	24.7	44.0	22.4	41.5	6.3	40	0.509

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	86.6	67.2	82.8	66.0	79.4	64.8	69.2	82.3	67.5	79.4	66.0	76.7	6.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.8	97.3	73.7	63.3	92.2	71.7	62.0	87.9	70.3	34.1	82.7	32.8	79.6	31.5	77.1	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.7	16.8	14.5	14.3	92.0	5.1	3.8	10.6	94.8	7.6	97.1	4.8	99.2	1.1	102.0		
(5)			13.5	72.3	5.1	1.5	9.8	73.3	6.8	74.2	3.9	75.0	0.2	76.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	34.4	36.3	43.0	50.0	55.6	63.4	66.9	65.2	59.4	51.6	42.4	36.3	(6)
(7)		DBStd	13.26	7.21	7.66	7.40	7.16	6.68	6.77	6.59	6.24	6.11	7.12	6.70	6.82	(7)
(8)		HDD50	1950	485	384	235	87	21	1	0	0	3	69	240	425	(8)
(9)		HDD65	5600	950	804	681	452	298	110	57	75	190	416	678	889	(9)
(10)		CDD50	2116	0	1	19	87	196	403	523	472	284	119	12	0	(10)
(11)		CDD65	288	0	0	0	1	8	62	115	81	21	0	0	0	(11)
(12)		CDH74	2311	0	0	0	9	58	499	982	627	133	3	0	0	(12)
(13)		CDH80	667	0	0	0	0	6	135	319	192	15	0	0	0	(13)
(14)	Wind	WSAvg	6.7	7.7	7.8	7.6	6.7	6.4	5.8	5.9	5.7	5.5	6.4	7.4	7.9	(14)
(15)	Precipitation	PrecAvg	36.0	3.1	2.6	2.4	2.5	3.3	2.6	3.2	2.9	2.6	3.5	3.7	3.5	(15)
(16)		PrecMax	45.0	7.4	5.1	6.8	7.0	5.8	4.7	6.7	6.4	5.6	6.7	10.7	9.4	(16)
(17)		PrecMin	25.4	0.7	0.2	0.0	0.3	0.7	1.1	0.4	1.1	0.6	1.0	1.1	0.3	(17)
(18)		PrecStd	5.1	1.5	1.4	1.7	1.6	1.3	1.0	1.3	1.4	1.5	1.8	2.1	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.9	56.5	65.7	75.3	80.3	89.5	92.6	90.9	82.3	73.1	60.0	52.6	(19)
(20)			MCWB	49.7	47.1	52.1	58.9	66.5	68.3	67.7	68.3	64.5	60.2	54.8	49.6	(20)
(21)		2%	DB	49.3	51.8	61.9	70.5	74.9	83.8	87.2	85.1	78.0	68.4	56.2	50.4	(21)
(22)			MCWB	47.4	46.0	50.2	56.4	61.8	66.9	66.7	67.0	63.9	59.5	52.7	48.5	(22)
(23)		5%	DB	46.6	48.8	57.7	66.2	71.6	79.7	83.4	80.8	74.1	64.5	53.8	47.9	(23)
(24)			MCWB	44.8	44.6	48.5	54.4	60.3	66.1	65.5	65.7	61.8	58.1	51.2	46.3	(24)
(25)		10%	DB	44.1	46.2	53.9	62.3	67.9	76.0	79.6	76.7	69.7	60.8	51.5	45.6	(25)
(26)			MCWB	42.7	43.0	46.7	52.2	58.3	64.1	64.1	64.0	59.7	56.4	49.7	44.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.5	49.1	54.0	60.4	67.7	71.7	70.3	70.9	67.5	62.9	56.2	50.7	(27)
(28)			MCD B	51.4	53.5	62.1	72.8	78.6	84.7	85.8	85.3	76.8	69.0	58.2	51.4	(28)
(29)		2%	WB	47.9	47.0	51.5	57.5	64.2	69.0	68.5	68.6	65.1	60.6	53.4	48.7	(29)
(30)			MCD B	49.1	50.5	58.5	68.3	72.8	79.9	82.8	81.8	74.3	66.2	55.3	50.0	(30)
(31)		5%	WB	45.0	45.2	49.7	55.1	61.4	66.8	67.0	66.8	63.1	58.7	51.6	46.5	(31)
(32)			MCD B	46.3	48.1	56.0	64.6	69.3	77.3	80.2	77.7	71.5	63.3	53.5	47.8	(32)
(33)		10%	WB	42.8	43.3	47.7	52.9	59.0	65.0	65.3	65.1	61.3	56.7	49.5	44.5	(33)
(34)			MCD B	44.0	45.9	53.1	60.9	65.9	74.4	77.3	74.5	68.3	60.2	51.3	45.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.0	10.1	13.9	16.0	15.9	17.5	17.7	16.8	15.9	12.7	8.9	7.8	(35)
(36)			MCDBR	9.9	14.3	19.6	21.3	21.0	22.5	23.3	23.0	21.5	16.9	11.6	9.8	(36)
(37)			MCWBR	8.4	9.7	11.4	11.7	11.6	10.3	9.2	9.5	10.1	9.6	8.4	8.2	(37)
(38)		5% WB	MCDBR	9.2	12.6	16.7	19.6	19.4	20.5	21.1	20.2	19.3	15.1	10.3	9.0	(38)
(39)			MCWBR	8.3	9.0	10.4	11.2	11.3	10.2	9.2	9.2	9.6	9.3	7.9	8.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.301	0.322	0.347	0.378	0.385	0.393	0.388	0.384	0.360	0.347	0.317	0.298	(40)	
(41)		taud	2.464	2.417	2.361	2.283	2.301	2.310	2.319	2.351	2.396	2.428	2.464	2.454	(41)	
(42)		Ebn at Noon	248	266	276	277	278	276	276	272	268	252	239	233	(42)	
(43)		Edh at Noon	20	26	33	39	40	40	39	36	31	26	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	310	566	961	1387	1628	1866	1777	1536	1189	720	369	251	(44)	
(45)		RadStd	44	93	129	185	182	176	156	145	103	54	36	45	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (49 neighbors)		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A

Nomenclature: See separate page